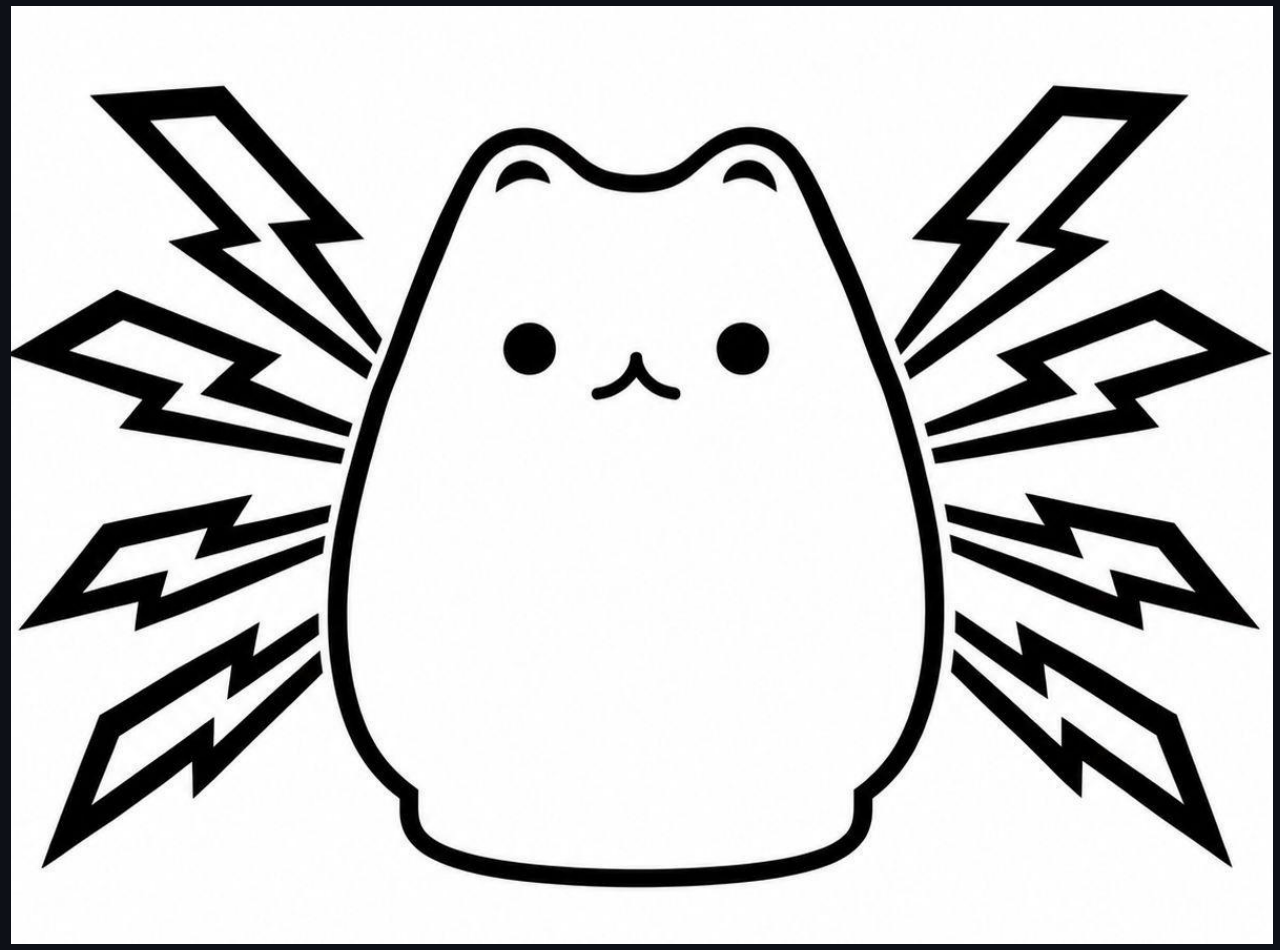
BSIDES NYC · HANDS-ON LAB

Learn Hardware Hacking With Your IoT Cat Lamp

The friendly smart hacker cat that grows with you.

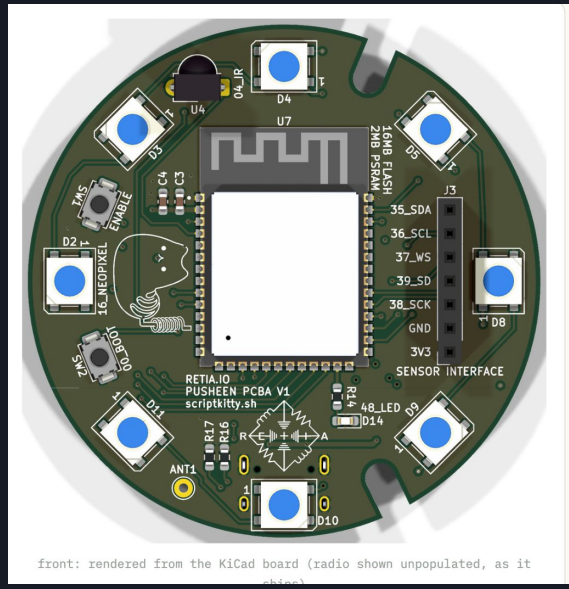
catlamp.site · scriptkitty.sh



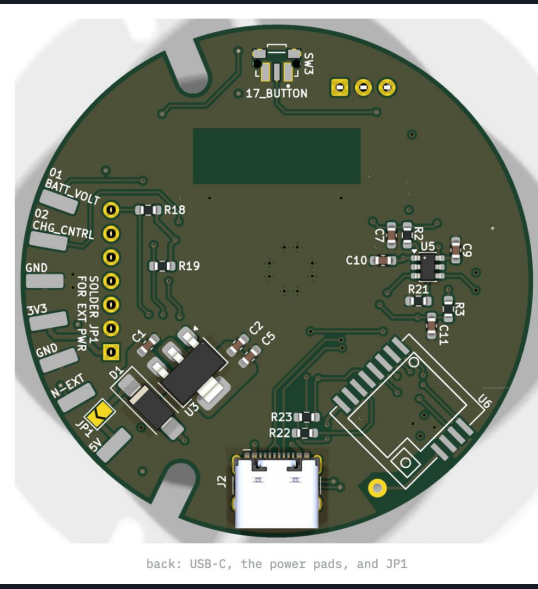
WHAT IS THIS?

A friendly Internet-of-Things lab for learning hardware hacking.

You get a real microcontroller in a cute case. We flash it, light it up, and then use it to poke at the devices and signals all around us.



front: rendered from the KiCad board (radio shown unpopulated, as it ships)



back: USB-C, the power pads, and JP1

retia.io · Pusheen PCBA v1

WHO AM I?

I’m Kody Kinzie

Security researcher focused on radio hacking, digital investigations, and cheap electronics. I love teaching beginners how to get started with ethical hacking.

Teaching
Script Kitty village · Cyber camps · Workshops · Custom hardware

Seen on
Null Byte · Hak5 · ScriptKitty · Retia · **hack.gay**



WHO IS SCRIPTKITTY?

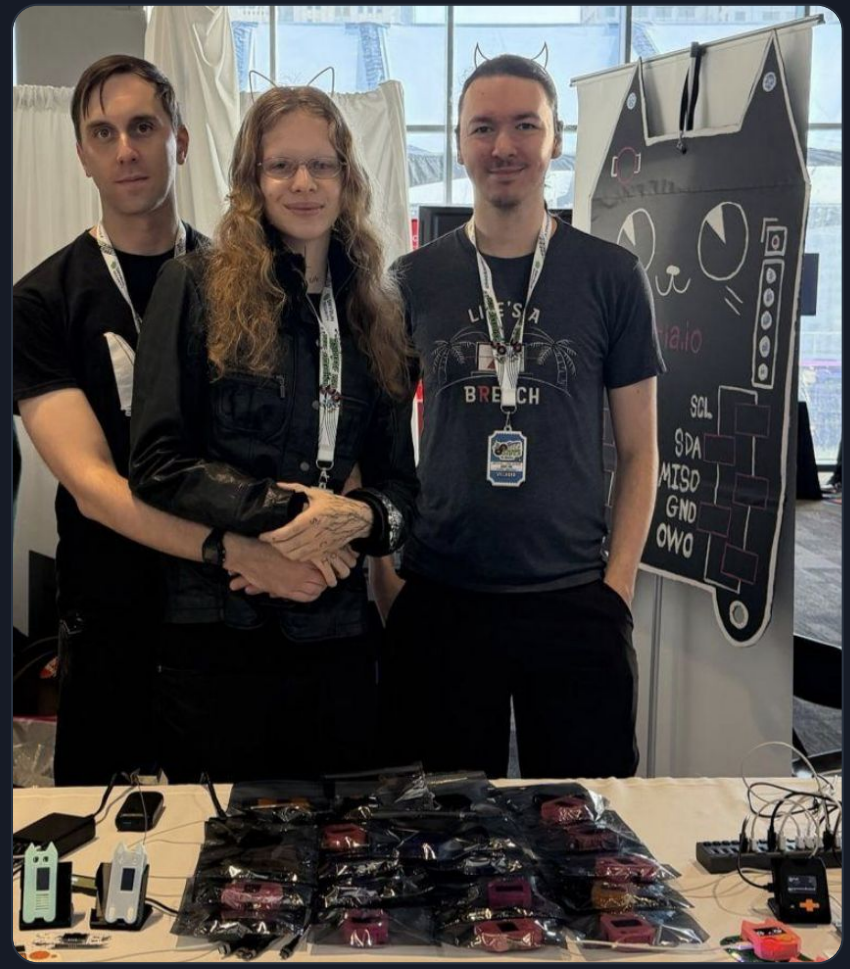
We’re Script Kitty Village

A community teaching ethical hacking with hands-on hardware — the same spirit as today’s lab.

Cyber camps & workshops · Custom hardware

Seen on Null Byte · Hak5 · Retia

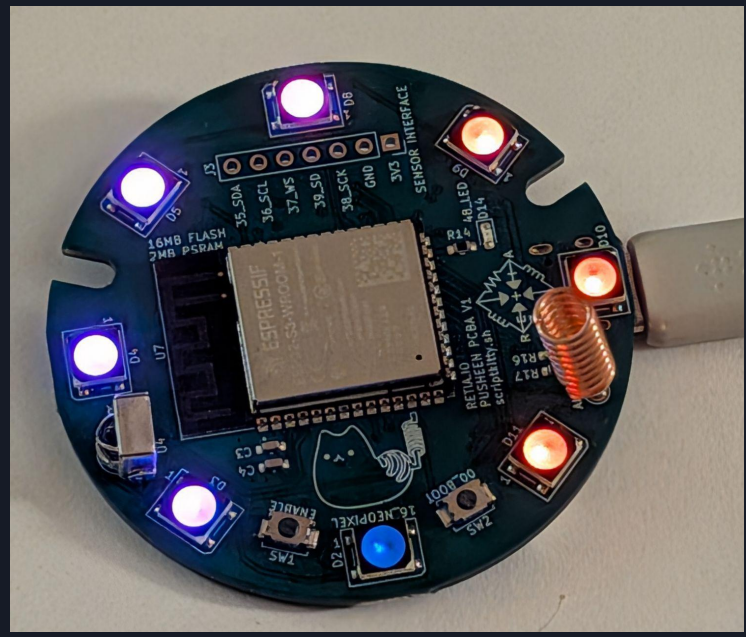
Find us at **DEF CON · HOPE · BSides · Chaos Congress**



THE HARDWARE

An ESP32-S3 brain with Wi-Fi & Bluetooth

- **Espressif ESP32-S3** — dual-core, native USB-C, Wi-Fi + BLE
- **8× addressable LEDs** — control one pixel at a time
- **Expandable** — mic, IR, LoRa radio, speaker, sensors
- **One small board** — becomes almost any IoT device



WHAT CAN IT BECOME?

What Can Your Cat Become?

Smart lamp
App & voice-controlled lighting

Smart speaker
Add an amp & speaker

Internet radio
Stream stations over Wi-Fi

Transcription
Mic in, text out

Wi-Fi scanner
See the networks around you

USB hacking tool
BadUSB keystroke injection

TODAY'S PLAN

Two of the most popular projects

01

WLED

The open-source smart light controller. Make the lamp glow and react to sound.

02

BitPirate

The hacker's brain. A terminal to drive LEDs, Bluetooth, Wi-Fi, and radio.



Along the way we'll learn how IoT devices really work — and start controlling hardware to **hack ethically**.

YOUR KIT

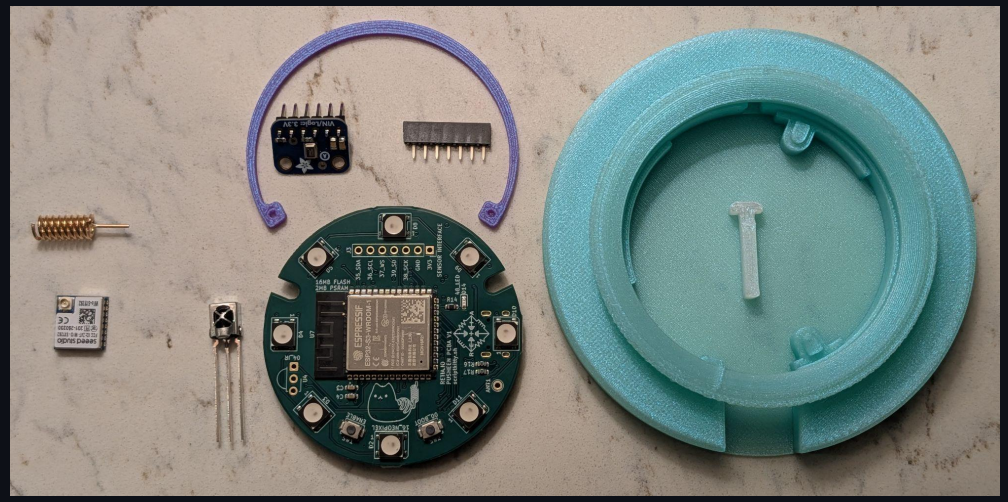
What’s in the box

In every kit

PCB · 3D-printed case · Push button · Retaining ring

Optional upgrades

IR receiver (color remote) · Microphone (audio-reactive) · $5 LoRa radio + antenna (mesh) · Amp & speaker (internet radio)



ASSEMBLE YOUR LAMP

Two steps to a finished case

STEP 1

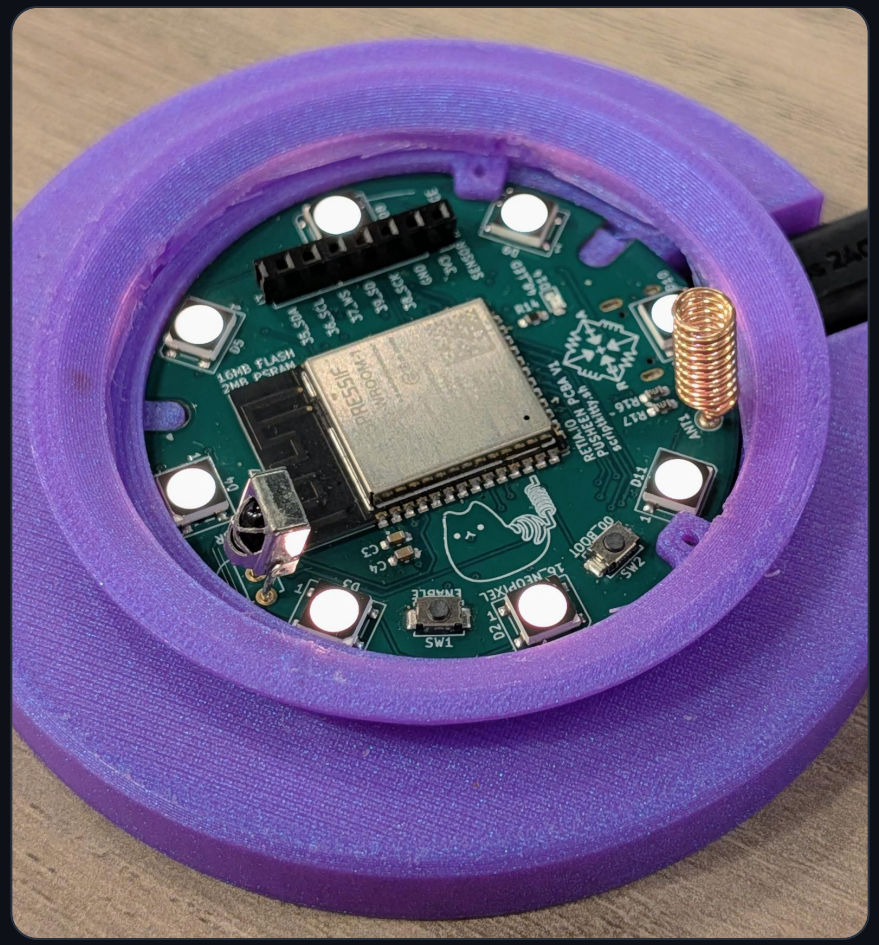
Insert the button

Work it back and forth until it slides smoothly. The notch faces up.

STEP 2

Secure it

Insert the puck, and drop the retaining ring into the slot to lock everything in place.



FIRST — A TRIP TO THE VET

Suspect trouble? Take your cat to the vet.

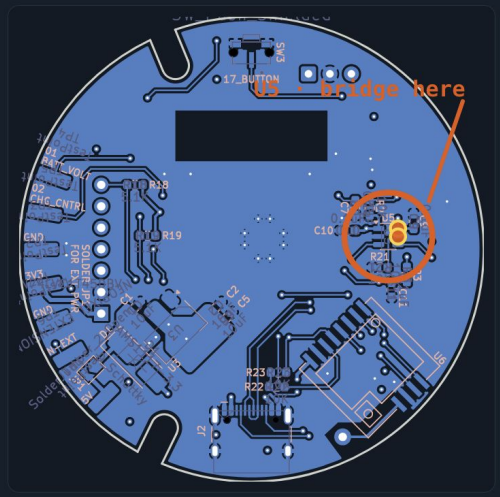
If a lamp won't light up or acts strange, the **vet scanner** checks the board and tells you exactly what's wrong — and how to fix it — before we flash.

```
https://scriptkitty.sh/vet.html
```

Bridge U5 pin 5 (DIR) to pin 6 (+5V)

The 8 NeoPixels are dark because the WS2812 level-shifter **U5 (SN74LVC1T45)** has its direction pin strapped the wrong way (R21 pulls DIR to GND → the translator runs backwards, ESP→LED never crosses). Tying DIR high fixes it. **U5 is on the BACK of the board.**

1 Find U5 — flip the board over



Back side, oriented like your photo: **SW3** at top, USB **J2** at bottom. U5 is the 6-pin chip on the right — **C10** to its left, **C9** to its right, **R21** just below.

2 Bridge these two pins



Pins 5 and 6 are **adjacent** on the C9 side — one solder blob does it. DIR is 5 V-tolerant, so tying it to +5 V reads HIGH and flips the shifter to ESP→LED.

Do it

1. Flip to the back; find **U5** (C10 left, C9 right, R21 below).
2. Flow a **solder bridge between pin 5 (DIR, middle) and pin 6 (top)** on the C9 side.
3. No reflash needed — the board is already driving GPIO16.

> **Prefer 3V3?** Bridge **pin 5 (DIR) → pin 1 (VCCA/+3V3)** instead — the diagonal top corners. Electrically cleaner, just a harder reach. Either works; you do not need to lift R21.

Confirm it worked

The puck plugged in now runs a bodge-checker. Watch the single **GPIO48 debug LED** on the front:

- **Flickering** + strip dark = still backwards
- **Steady on** + 8 LEDs rainbow = **fixed**

> The moment the bridge takes, both change together. Then say the word and I'll reflash WLEDkitty.

HOW FLASHING WORKS

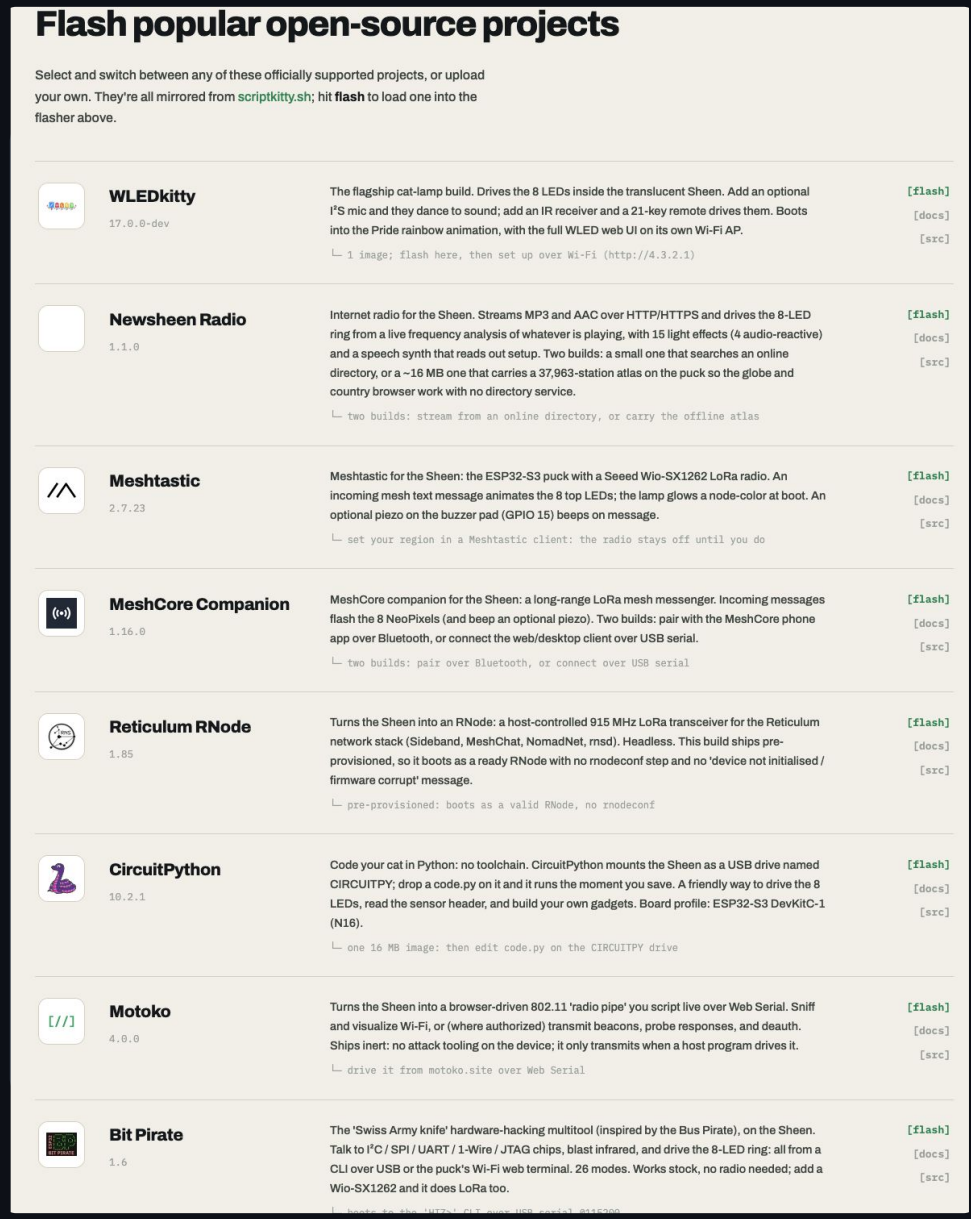
One program at a time

IoT devices are too **smol** to run more than one program. To switch what the lamp *is*, flash a new project onto it — from the **browser** or over **Wi-Fi**.

Flash in the browser
Plug in USB-C, pick a project, click flash

Switch any time
Re-flash to swap projects in seconds

Try your first one at **catlamp.site**



WHERE TO FLASH

Two websites, one cat

They overlap — but **scriptkitty.sh** is the advanced flasher with every project and more features, while **catlamp.site** is the simple one-click flow just for the Sheen.

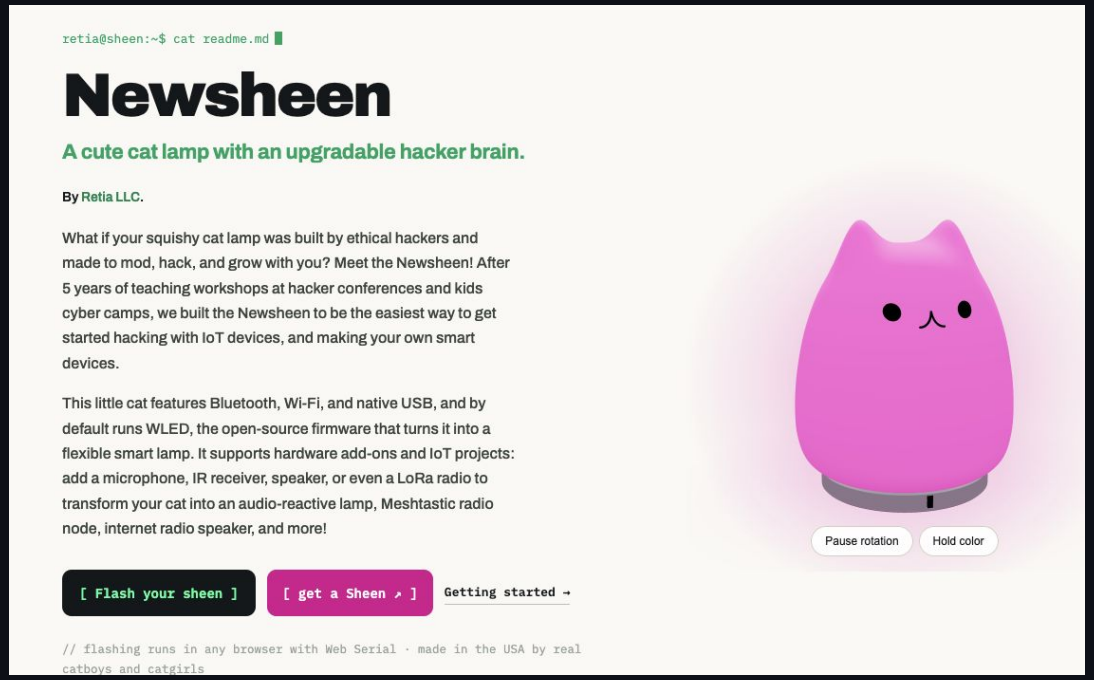
scriptkitty.sh

Every ScriptKitty project · more advanced · upload your own



catlamp.site

Just the Sheen · simplest one-click flow for beginners



PROJECT 01

WLED

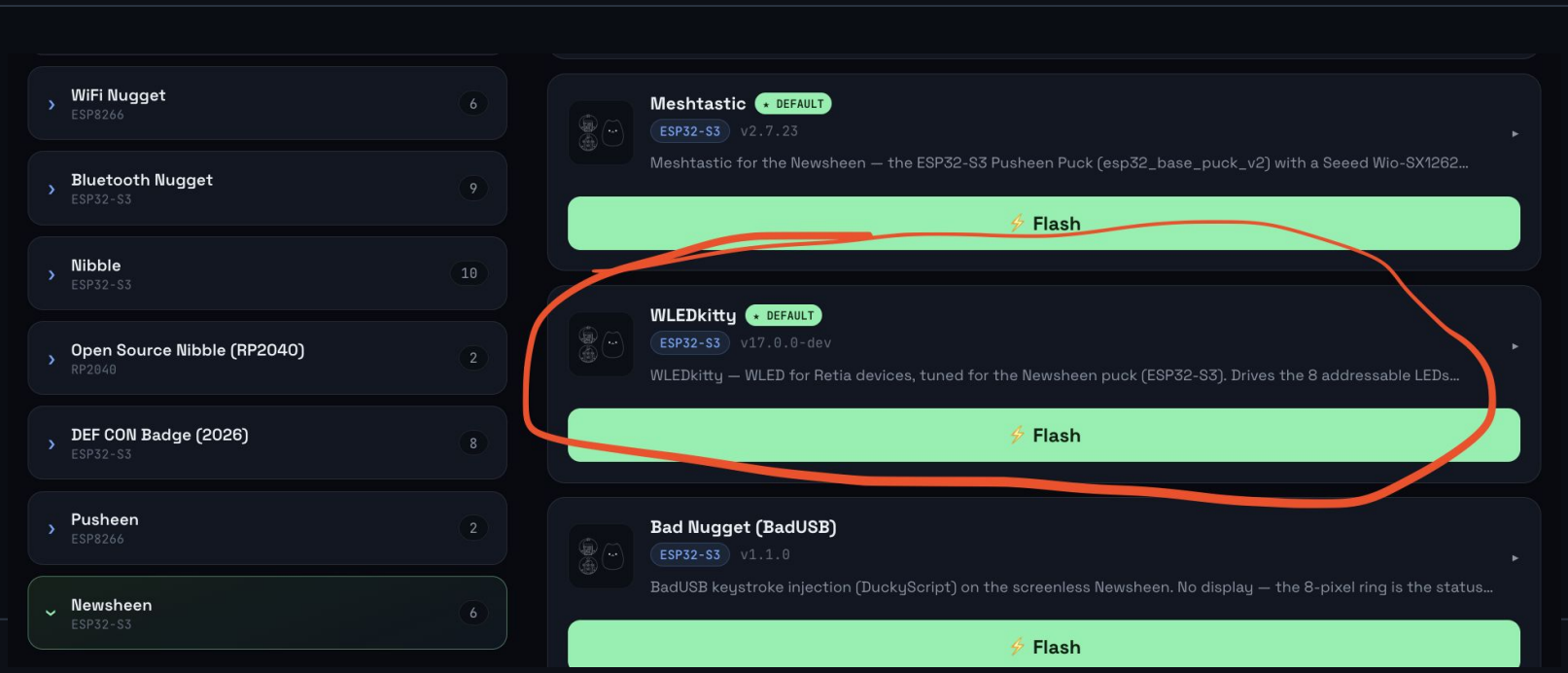
The open-source smart light controller that works over Wi-Fi.



STEP 1 — FLASH IT

Install WLED from the browser

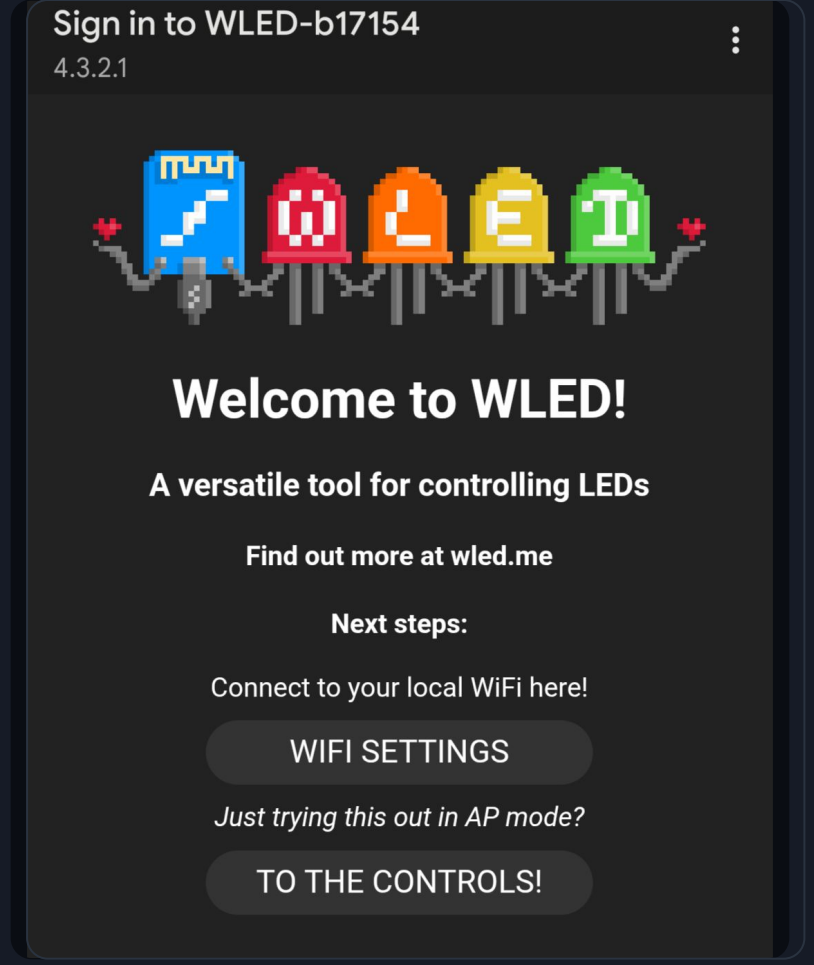
1. Open **scriptkitty.sh** and plug in your lamp with USB-C
2. Find **WLEDkitty** and click its **Flash** button
3. Wait for it to finish — about 30 seconds

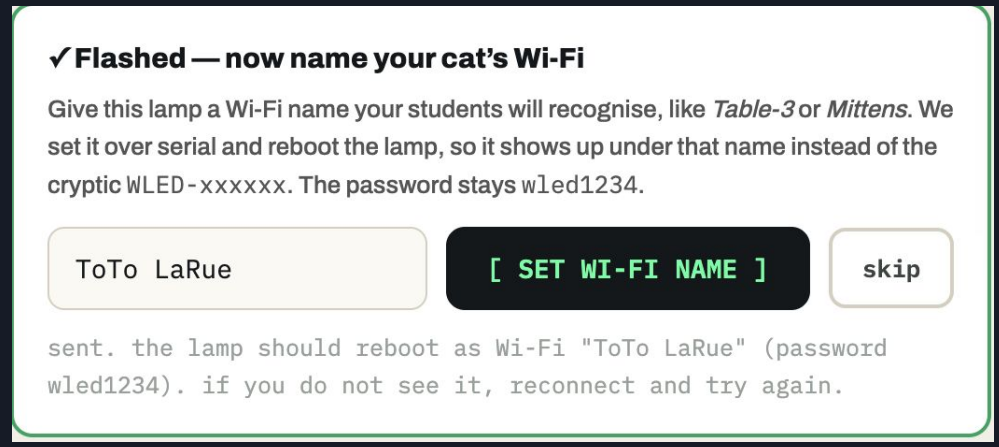


STEP 2 — IT'S ALIVE!

Power it up & connect

- · Plug in — wait a second for the LEDs to glow (no light? replug)
- · Join Wi-Fi **WLED-AP** · password **wled1234**
- · Hold your phone **close to your PCB** — lots of these in the room
- · No sign-in page? Browse to **4.3.2.1**
- · Change the color to confirm you've got **your** lamp





TRY IT

Animations & remote control

Change the animation

Open the effects list and try a few — tweak speed, intensity, and palette.

Click the button for soft on/off

Turn the lamp on and off with the 3d printed button



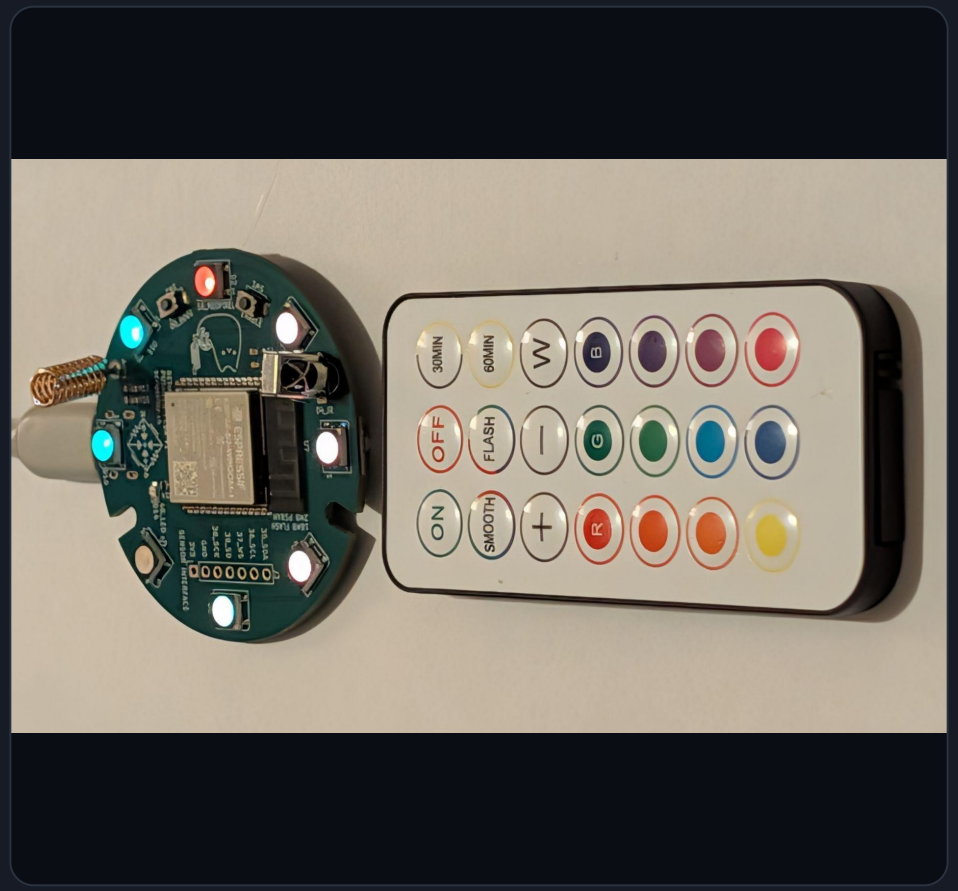
HARDWARE UPGRADES

Small add-ons, big new powers

IR receiver
Control the lamp with a common color remote

Microphone
Lights go audio-reactive to music

LoRa radio
Off-grid mesh with Meshtastic & Meshcore



PROJECT 02

BitPirate

The hacker’s brain for the cat: a terminal that gives you full control of the hardware.



WHAT ELSE CAN IT DO?

Beyond what we'll cover today

Extract firmware, read sensors, and talk to other hardware over **UART, serial, infrared, I²C, SPI, JTAG**, and more.

Pick a dedicated adapter and the cat speaks the languages electronics use to talk to each other.



GET STARTED

Flash BitPirate & open the terminal

1. Flash **BitPirate** at **scriptkitty.sh**
2. Click **Connect** and select your board
3. Type **help**




```
=== Help Shell: Commands by Mode ===

Select a mode to get help:
  [1] *  ALL
  [2]    1WIRE
  [3]    UART
  [4]    HDUART
  [5]    I2C
  [6]    SPI
  [7]    2WIRE
  [8]    3WIRE
  [9]    DIO
  [10]   LED
  [11]   INFRARED
  [12]   USB
  [13]   BLUETOOTH
  [14]   WIFI
  [15]   JTAG
  [16]   I2S
  [17]   CAN
  [18]   ETHERNET
  [19]   SUBGHZ
  [20]   RFID
  [21]   RF24
  [22]   FM
  [23]   CELL
  [24]   LORA
  [25]   EXPANDER
  [26]   Exit
Enter index (default 1):
```

MODE: LED

Control the LEDs by hand

Switch to the LED module, then drive each pixel yourself:

- · **mode led** — enter LED control
- · **set 0 red** — color the first pixel
- · **set 1 blue** …and so on down the strip

Manual control of the LEDs — try it!



MODE: USB

A wired mouse jiggler

In USB mode the cat becomes a **USB HID device** — plug it into a computer and it can act as a mouse. Jiggle the cursor to keep a screen awake, straight over the cable, no pairing needed.

```
cat@lamp:~$ mode usb
USB> mouse jiggle 100 # every 100 ms
```

```
USB> » mode usb
USB> mode usb
Mode changed to USB

USB> » mouse jiggle 100
USB> mouse jiggle 100
USB Mouse: Configuring HID...
USB Mouse: Initialize HID...
USB Mouse: Jiggle started (1000 ms)... Press [ENTER] to stop.
```

MODE: BLUETOOTH

Scan, sniff, and pair

Switch to Bluetooth and the cat can see the BLE devices around it — and act as one itself.

```
cat@lamp
:~$
mode bluetooth
```

```
====================================
BLUETOOTH
====================================

scan                    - Discover devices
pair <mac>              - Pair with a device
sniff                   - Sniff Bluetooth data
spoof <mac>             - Spoof mac address
status                  - Show current status
server                  - Create an HID server
keyboard [text]         - Start keyboard bridge
mouse [action]          - Move and click mouse
mouse jiggle [ms]       - Random mouse moves
reset                   - Reset interface
```

LIVE DEMO

A Bluetooth mouse jiggler

Start a BLE HID **server**, pair to the cat from your laptop, then jiggle the mouse to keep a screen awake.

```
==================================
BLUETOOTH
==================================

scan                - Discover devices
pair <mac>          - Pair with a device
sniff               - Sniff Bluetooth data
spoof <mac>         - Spoof mac address
status              - Show current status
server              - Create an HID server
keyboard [text]     - Start keyboard bridge
mouse [action]      - Move and click mouse
mouse jiggle [ms]   - Random mouse moves
reset               - Reset interface

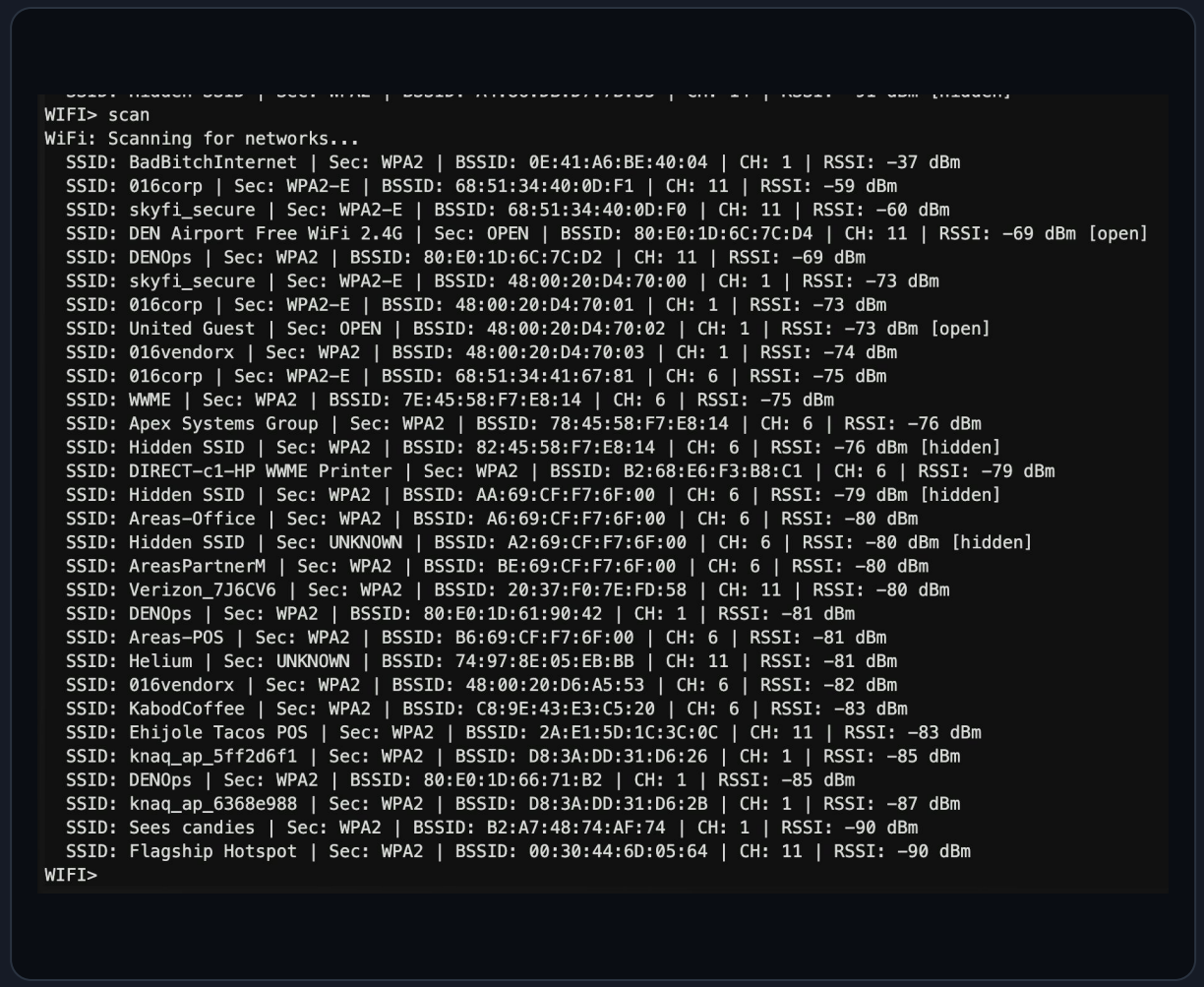
BLUETOOTH> » server
BLUETOOTH> server
Bluetooth Server: Starting BLE HID server as "Bit-Pirate-Bluetooth"...
→ You can now pair from your phone or computer.
BLUETOOTH> » mouse jiggle 100
BLUETOOTH> mouse jiggle 100
Bluetooth Mouse: Jiggle started (100 ms)... Press [ENTER] to stop.
```

MODE: WIFI

See a network inside & out

Switch to Wi-Fi mode, then scan to see what's around you and join a network to look at it from within.

- · **mode wifi** — switch to Wi-Fi mode
- · **scan** — what's out there
- · **connect <ssid> <password>** — join it
- · **ping** · **nmap** — explore from within



MODE: WIFI

Connect to a network

Join a network with **connect <ssid> <password>**. Once you're on, the cat can run network commands from the inside.

```
WIFI> connect MyWiFi mypassword
✓ Connected successfully.
```

Connected over serial? Reset and pick **Wi-Fi Connect** to open the on-board Web UI instead.



MODE: WIFI · CONNECTED

Explore from the inside

Now that you're on the network, poke around:

- · **discovery** — find other devices on the network
- · **http get <url>** — fetch a web page
- · **ping <host>** — check a host is alive
- · **nmap <host>** — scan its open ports

```
WIFI> » http get google.com
WIFI> http get google.com
HTTP: Sending GET request to https://google.com...

========== HTTP GET =============
HTTP/1.1 301
Location: https://www.google.com/
Server: gws
WIFI> » nmap 148.72.245.204
WIFI> nmap 148.72.245.204
Nmap: Using top 100 common ports (may take a few seconds)
»
Nmap scan report for 148.72.245.204 (148.72.245.204)
Host is up (986ms latency).
PORT       STATE          SERVICE
13/tcp     filtered       daytime
21/tcp     filtered       ftp
22/tcp     open           ssh
25/tcp     filtered       smtp
26/tcp     filtered       rsftp
37/tcp     filtered       time
79/tcp     filtered       finger
80/tcp     filtered       http
81/tcp     filtered       hosts2-ns
```

MODE: WIFI

Share your connection with repeater

Run **repeater** and the cat rebroadcasts the Wi-Fi it's connected to as a brand-new network — **extending the range** and letting others hop online through your uplink.

- · Pick a new **SSID** and **password** for the repeated network
- · Devices join it at **192.168.4.1**, up to 10 at once
- · **repeater stop** ends it

```
WIFI> » repeater
WIFI> repeater

WiFi Repeater: Forwarding traffic from uplink.
Enter Repeater SSID [esp32repeater]: » BadBitchInternet69
BadBitchInternet69
Enter Repeater Pass [esp32bitpirate]: » password1234
password1234

WiFi Repeater: Starting repeater...

WiFi Repeater: Routing traffic between uplink and repeater...

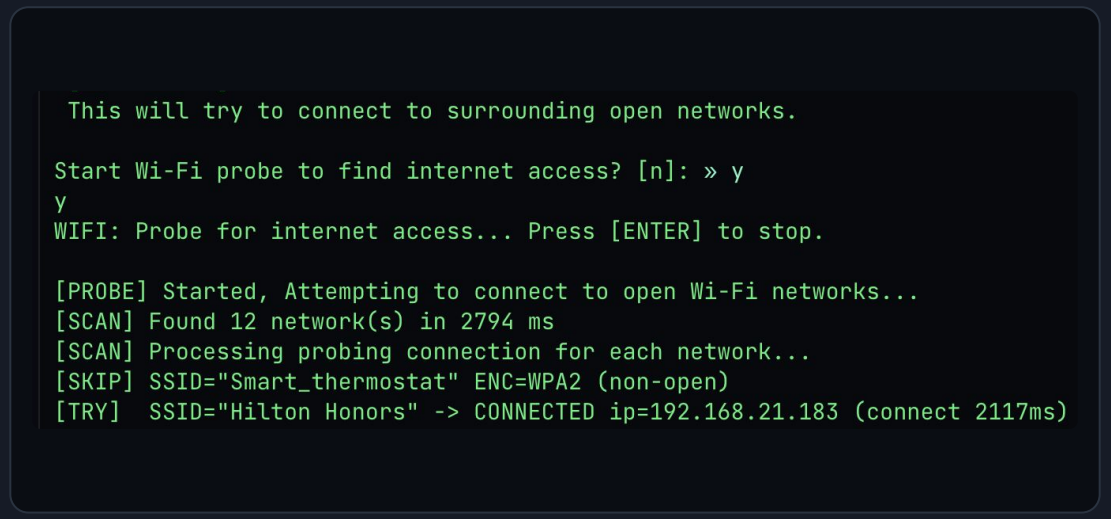
  Uplink           : BadBitchInternet
  Repeater SSID    : BadBitchInternet69
  Repeater Pass    : pa********4
  Repeater IP      : 192.168.4.1
  Max connections  : 10

  Use 'repeater stop' to stop.
```

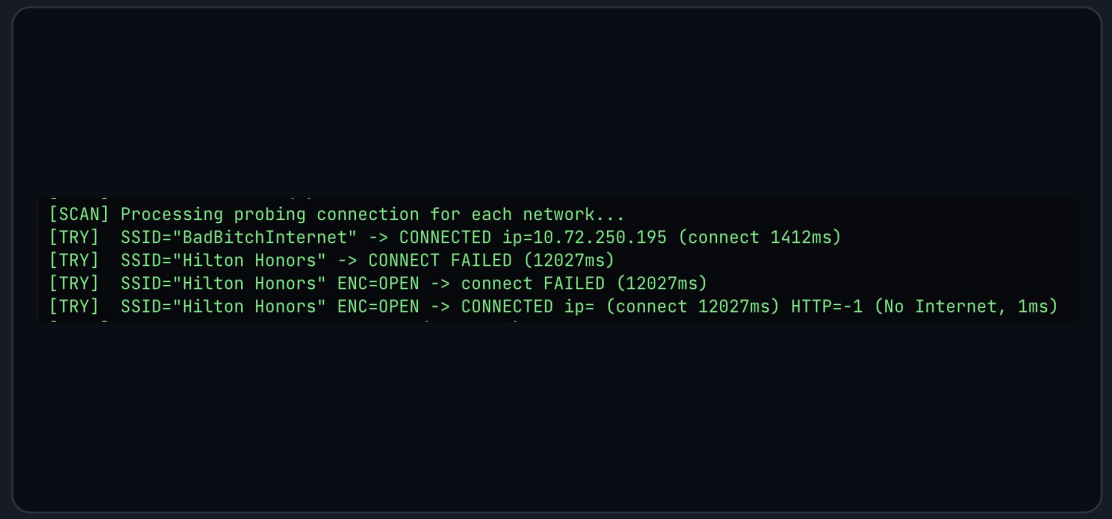
MODE: WIFI

Hunt for open internet with probe

Run **probe** and the cat tries to connect to **every open Wi-Fi network** around you, one by one — then tells you whether each actually reached the internet: a **connect / ping time** if it worked, or **No Internet** if it went nowhere.



It skips secured networks and **connects** to open ones, showing the time



A captive portal with no real access reports **No Internet**

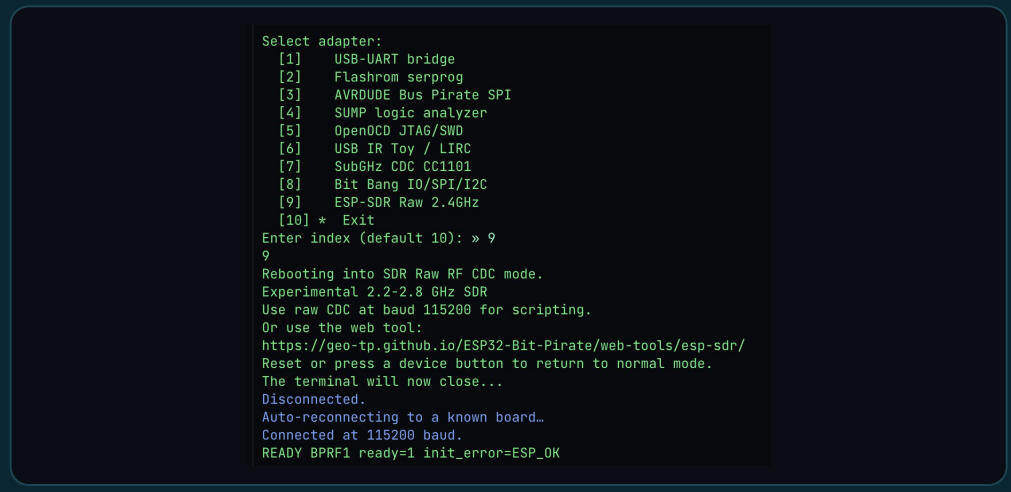
MODE: USB - SDR · BRAND NEW

Your cat is a software-defined radio

Hidden in the USB adapters is a raw 2.4 GHz SDR — added just **10 days ago**. Switch into it in three steps:

```
cat@lamp:~$ mode usb
USB> adapters
select > 9 # ESP-SDR Raw 2.4GHz
```





SDR · WEB INTERFACE

Watch the airwaves in your browser

Once it reboots into SDR mode, open the web tool and hit **connect**. Adjust the window to watch transmissions appear — Wi-Fi and Bluetooth light up the 2.4 GHz band.

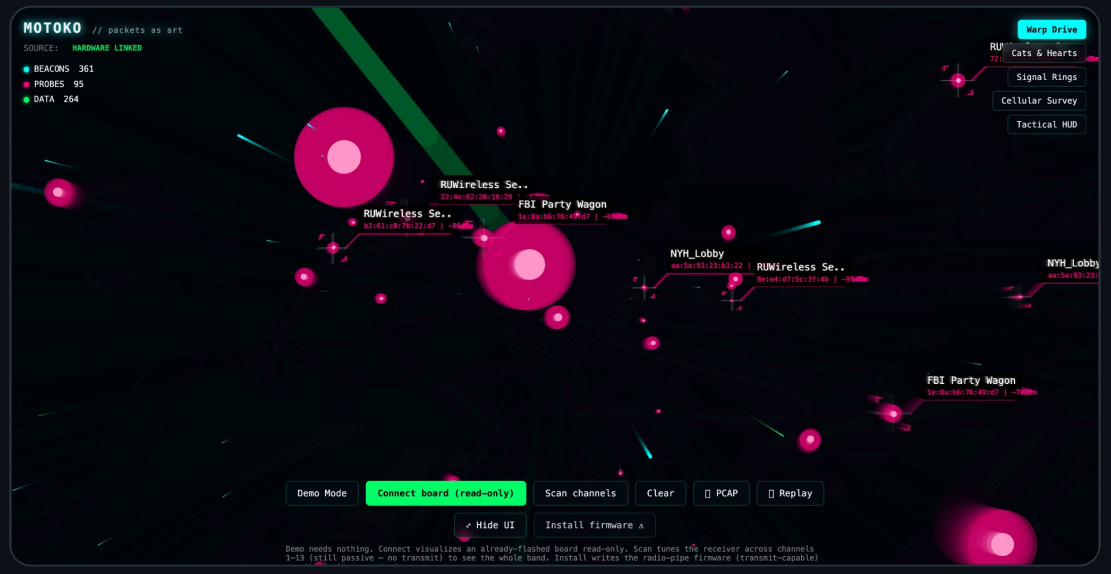
geo-tp.github.io/ESP32-Bit-Pirate/web-tools/esp-sdr/



GO FURTHER

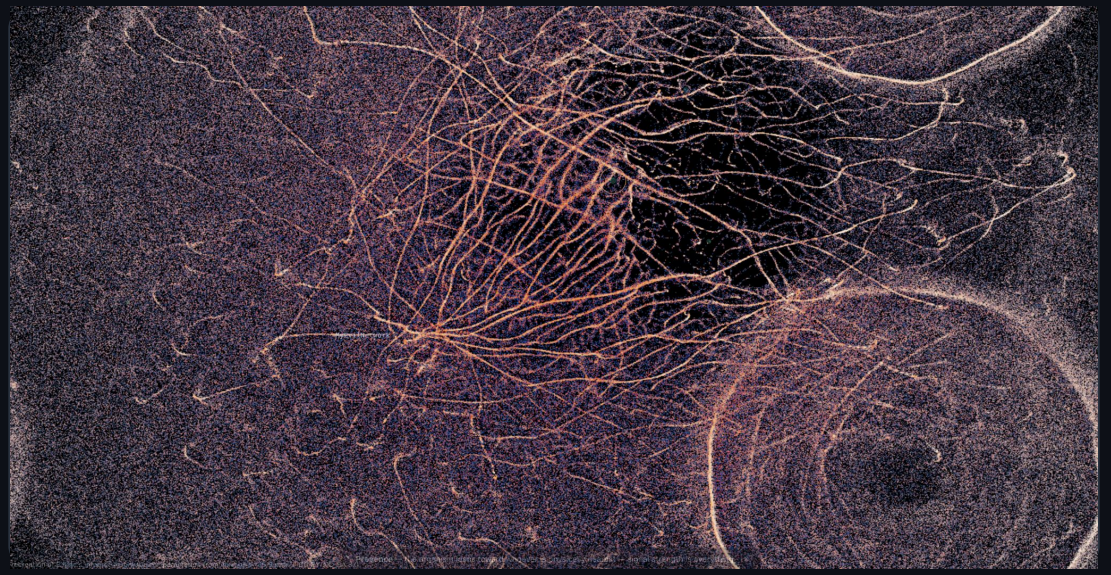
Take it home & keep playing

Control it over Wi-Fi with no cable, then point the same packet-sniffing skills at bigger, stranger, more beautiful things.



motoko.site

Visualize the Wi-Fi world around you



crimes.gay

Art that reacts to the devices nearby

THAT'S A WRAP

Thanks for building with us!

You flashed firmware, lit up LEDs by hand, and turned a cat into a Bluetooth, Wi-Fi, and radio tool.

retia.io · catlamp.site · scriptkitty.sh

